

Department of Electrical and Electronic Engineering

Project Report

Course Title : Electronics II

Course Code : EEE225.3

Project Name : Solar Panel Sun-Tracking System

Group members

MD. ZISAN AL ZAHAN

ABDULLAH AL KAFI

MD. TALHA TOUHID MAHIM

Submitted to : Dr. Md. Shahjalal

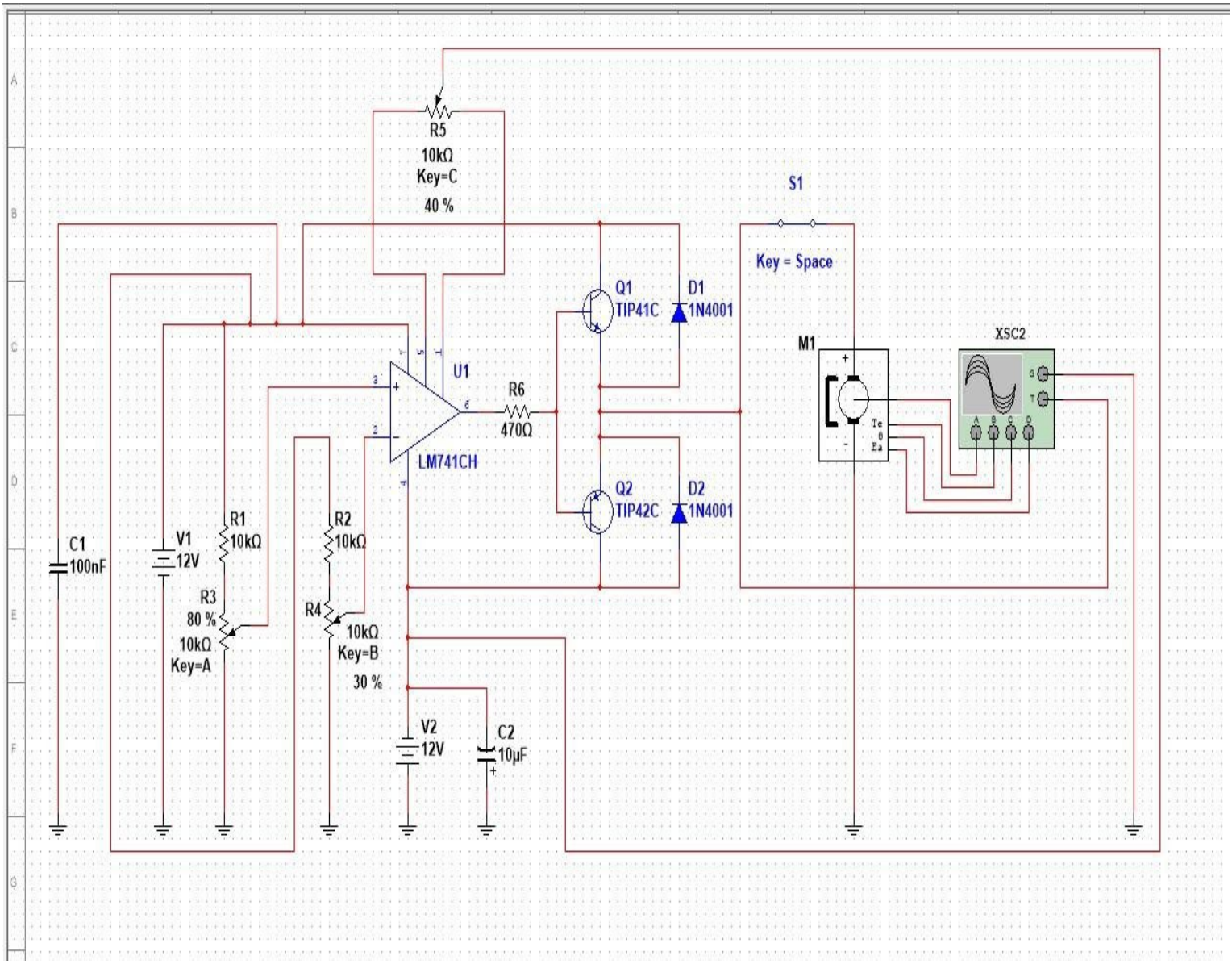
1.

Submission Date : 10/06/2026

Project Overview:

The primary objective of this project is to design and simulate a single-axis solar tracking system intended to automatically align a solar panel with the sun's trajectory, thereby maximizing energy absorption. The system employs an LM741CH operational amplifier acting as a comparator to process signals derived from light sensors, which are simulated using potentiometers. The system subsequently drives a DC motor through a push-pull transistor amplifier to physically adjust the optimal angle of the panel.

Circuit Diagram:



2.

Component Inventory:

The circuit consists of the following components integrated within the NI Multisim environment:

Component Name	Quantity	Designator
LM741CH Operational Amplifier	01	U1
TIP41C NPN Power Transistor	01	Q1
TIP42C PNP Power Transistor	01	Q2
10 KΩ Potentiometers	02	R3, R4
10 KΩ Resistors	02	R1, R2
470 Ω Resistor	01	R6
10 KΩ Potentiometer	01	R5
1N4001 Diodes	02	D1, D2
100 nF Capacitor	01	C1
10 µF Capacitor (+)	01	C2
DC Voltage Sources	02	V1, V2
DC Motor	01	M1
SPST Switch	01	S1
4-Channel Oscilloscope	01	XSC2
Ground	08	G

3.

Simulation Verification & Performance Analysis:

The operational success of the simulation is validated by continuously monitoring the dynamic interactions between the sensor inputs (potentiometers) and the resulting motor behavior:

- **Balanced State (Sun Centered):** When both potentiometers (R3 and R4) are configured to equivalent percentage values, the voltage differential between Pin 2 and Pin 3 of the LM741CH operational amplifier is 0V.
- **Balanced State (Output):** The Op-Amp output remains at 0V, ensuring both transistors remain in the off state.
- **Balanced State (Motor):** The motor remains completely stationary, indicating perfect mechanical alignment with the simulated sun.
- **Unbalanced State (Sun Shifting East/West):** Modifying the resistance of R3 (Key=A) or R4 (Key=B) simulates the sun moving into closer proximity to one of the sensors, creating a voltage difference.
- **Forward Rotation (Eastward):** If the non-inverting input voltage surpasses the inverting input voltage, the LM741CH output is driven to the positive saturation rail, activating the NPN transistor (Q1) and rotating the motor forward (Eastward).
- **Reverse Rotation (Westward):** If the inverting input voltage is higher, the Op-Amp produces a negative saturation voltage, activating the PNP transistor (Q2) and reversing the motor direction to backward (Westward).
- **System Correction:** The simulation demonstrates accurate corrective behavior, as altering a potentiometer value compels the motor to operate continuously until a state of equilibrium is achieved.

4. Oscilloscope Observations and Diagnostic Routing

To conduct a comprehensive analysis of the mechanical and electrical characteristics of the tracking system, a 4-channel oscilloscope (XSC2) is interfaced with specific monitoring pins on the DC motor (M1):

- **Channel A:** Interfaced with the middle pin of the motor, representing the mechanical shaft output.
- **Channel B:** Interfaced with the Te Pin, corresponding to Electromagnetic Torque.
- **Channel C:** Interfaced with the θ Pin, denoting Angular Position.
- **Channel D:** Interfaced with the Ea Pin, representing Armature Voltage/Back-EMF, which directly correlates to the motor's RPM.

5. Team Contributions

The project architecture was divided into three core domains, with responsibilities allocated as follows:

Part 1: Component Selection & Power Stage Design

Prepared by: MD. TALHA TOUHID MAHIM

- **Objective:** Designing the power delivery and motor driving architecture of the circuit.
- **Implementation:** As standard operational amplifiers are insufficient for sourcing or sinking the current required to drive a DC motor directly, a power amplification stage was designed.
- **Push-Pull Amplifier:** A push-pull amplifier was integrated using complementary bipolar junction transistors: the TIP41C (NPN) and TIP42C (PNP).
- **Current Handling:** This configuration efficiently handles the bidirectional current required to rotate the motor clockwise and counter-clockwise.
- **Power Supply:** The circuit is energized by a dual ±12V DC supply (V1 and V2), permitting the Op-Amp to swing both positively and negatively for bidirectional control.
- **Filtering:** Filtering capacitors (C1 and C2) were utilized to stabilize the supply.
- **Protection:** To mitigate sudden voltage spikes generated by the inductive load of the DC motor upon deactivation or reversal, two 1N4001 flyback diodes (D1 and D2) were placed across the motor driver outputs for transistor protection.

Part 2: Control Logic & Sensor Integration

Prepared by: ABDULLAH AL KAFI

- **Objective:** Designing the core control logic, sensor inputs, and comparator circuit to dictate motor actuation.
- **Implementation:** The LM741CH (U1) operational amplifier was configured as a voltage comparator to continuously evaluate voltage levels at its inverting (Pin 2) and non-inverting (Pin 3) inputs.
- **Sensors:** Physical Light Dependent Resistors (LDRs) were simulated utilizing 10 KΩ potentiometers (R3 and R4).
- **Offset Correction:** An additional 10 KΩ potentiometer (R5) was integrated between pins 1 and 5 to successfully mitigate offset error.
- **Simulation Mechanics:** By adjusting the keys, changing light intensity can be mimicked as the sun moves across the sky.

- **Logic Execution:** When one sensor receives more light, its resistance decreases, shifting the input voltage.
- **Transistor Activation:** The LM741CH outputs a corresponding positive or negative signal that activates either the TIP41C or TIP42C transistor.
- **Target Balance:** This drives the motor to correct the panel's position until light distribution is balanced, neutralizing the voltage difference and halting the motor.

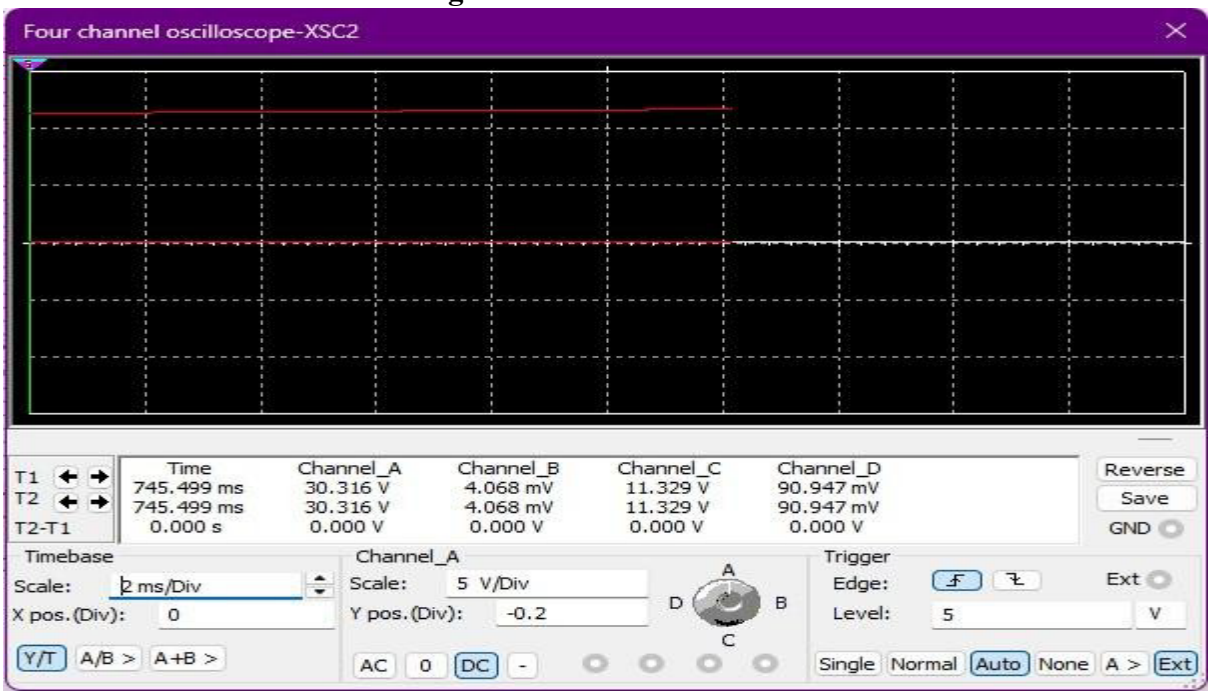
Part 3: System Simulation & Oscilloscope Analysis

Prepared by: Zisan Al Zahan

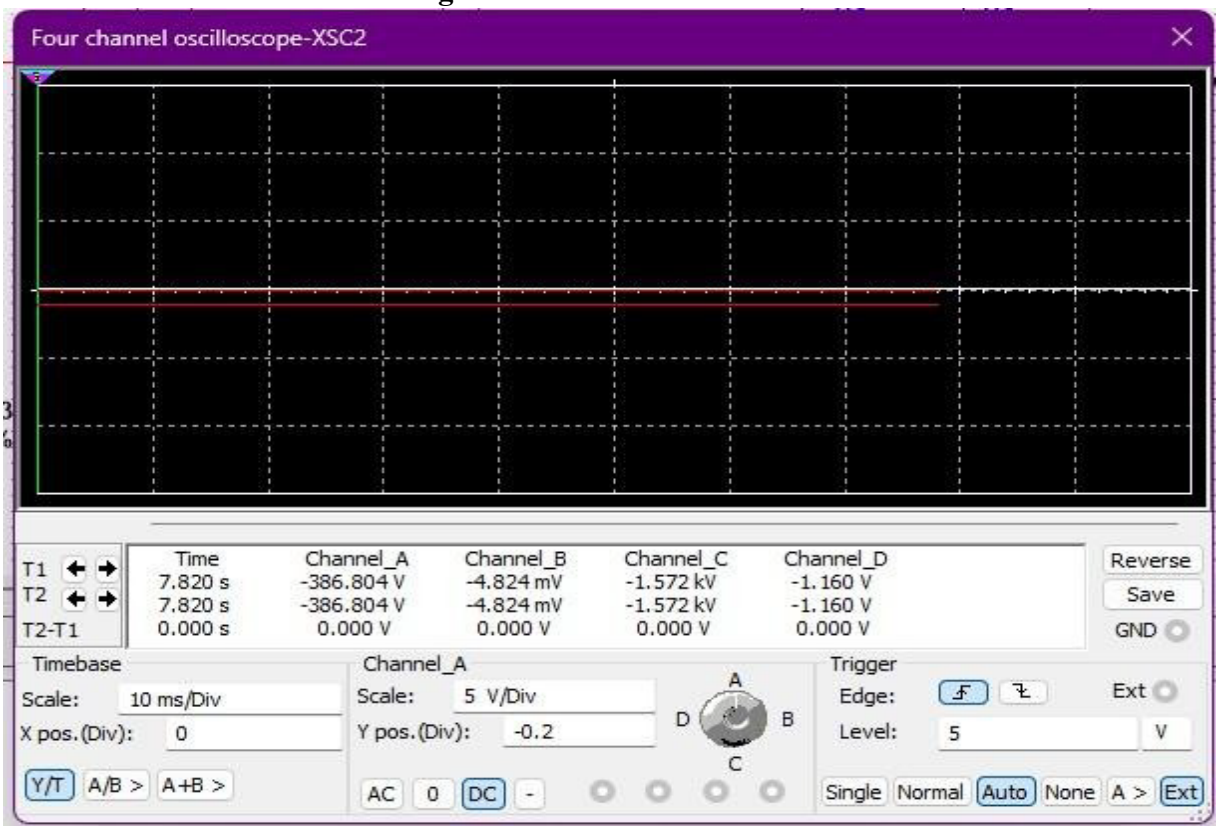
- **Objective:** Validating theoretical circuit functionality via NI Multisim simulation and extracting rigorous performance data.
- **Implementation:** Executed the final schematic routing, ensuring all ground connections were secured, and incorporated a master SPST switch (S1, Key=Space) for motor power testing.
- **Oscilloscope Configuration:** The 4-channel oscilloscope (XSC2) was precisely wired to the DC motor's specialized output pins to monitor performance metrics, linking Channel A to the middle pin, Channel B to the Torque pin (Te), Channel C to the angular pin (θ), and Channel D to the RPM output pin (Ea).
- **Eastward Tracking Results:** Simulation confirmed that when the solar panel tracks Eastward, the graph displays on the positive axis, and all values across channels A, B, C, and D appear positive.

  Westward Tracking Results: Conversely, when the solar panel tracks Westward, the graph transitions to the negative axis, and all values across channels A, B, C, and D appear negative.

When the Solar Panel is following the sun Eastward:



When the Solar Panel is following the sun Westward:



Validation: Modulating the potentiometer values successfully demonstrated smooth forward and reverse motor engagement without Op-Amp overload, ensuring high-fidelity data waveforms for the final analysis.

6. Conclusion

Through the collaborative execution of power stage design, comparator logic implementation, and rigorous Multisim analysis, the team successfully developed a fully functional simulation of a Solar Panel Sun Tracking System. The resulting circuit precisely measures simulated light differentials and accurately drives the motor bidirectionally, establishing the viability of this architectural design for real-world solar energy optimization applications.

Team Contributions Summary:

MD. TALHA TOUHID MAHIM	————	Component Selection & Power Stage Design
ABDULLAH AL KAFI	————	Control Logic & Sensor Integration
MD. ZISAN AL JAHAN	————	System Simulation & Oscilloscope Analysis